

Extension to Pension Run

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1. Introduction

2. Institutional Background

3. Model

The time is continuous with an infinite horizon. The framework follows [Achdou, Han, Lasry, Lions, and Moll \(2022\)](#), which is a continuous time analogue of the Aiyagari–Bewley–Huggett model ([Aiyagari \(1994\)](#); [Huggett \(1993\)](#)). We integrate the model with the standard life-cycle model as in [Huggett \(1996\)](#). In the model, agents have different believes in the possible reforms.

3.1. Demographics

Let b be the fertility rate and $m(s)$ the mortality rate. We assume that the individuals cannot live longer than T and $n(t, T) = 0$. Let $n(t, s)$ be the unnormalized population density. The population dynamic is given by

$$\frac{\partial n}{\partial t}(t, s) = -\frac{\partial n}{\partial s}(t, s) - m(s)n(t, s), \quad n(t, 0) = b \int_0^T n(t, s)ds. \quad (1)$$

Let $N_t = \int_0^T n(t, s)ds$ be the total population. Detrending the population equation by $\hat{n}_t = n_t/N_t$ gives

$$\frac{\partial \hat{n}}{\partial t}(t, s) = -\left(\frac{\dot{N}(t)}{N(t)} + m(s)\right)\hat{n}(t, s) - \frac{\partial \hat{n}}{\partial s}(t, s), \quad \hat{n}(t, 0) = b. \quad (2)$$

The population growth rate η_t is given by

$$\begin{aligned} \frac{\dot{N}_t}{N_t} &= \frac{\int_0^T \frac{\partial n}{\partial t} ds}{\int_0^T n(t, s)ds} = \frac{\int_0^T -\frac{\partial n}{\partial s}(t, s) - m(s)n(t, s)ds}{\int_0^T n(t, s)ds} \\ &= \frac{-(n(t, T) - n(t, 0)) - \int_0^T m(s)n(t, s)ds}{\int_0^T n(t, s)ds} = b - \int_0^T m(s)\hat{n}(t, s)ds \end{aligned} \quad (3)$$

In the steady state, the population grows at a constant rate $\eta_t = \frac{\dot{N}(t)}{N(t)}$. The steady state $\hat{n}$ is given by

$$\bar{n}(s) = b \exp\left(-\int_0^s \eta + m(x)dx\right). \quad (4)$$

The normalization condition $\int_0^T \bar{n}(s)ds = 1$ determines η :

$$b \int_0^T \exp\left(-\int_0^s \eta + m(x)dx\right)ds = 1. \quad (5)$$

3.2. Household

Retirees. Given the interest rate r , a retiree that dose not participate the monthly plan maximizes

$$V_t^\ell = \max \int_t^T e^{-\int_t^s \rho + m(x)dx} [u(c_s) + m(s)b(a_s)] ds \quad (6)$$

subject to

$$\dot{a}_t = r_t a_t + \text{tr}_t - c_t \quad (7)$$

$$a_t \geq 0. \quad (8)$$

The utility flow is given by

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma} \quad (9)$$

and the bequest motive follows [De Nardi \(2004\)](#) and [French and Jones \(2011\)](#):

$$b(a) = \theta_b \frac{(a + \kappa)^{1-\gamma}}{1-\gamma}. \quad (10)$$

We assume that the transfer tr_t is a flow from the bequest, which is assumed to be distributed uniformly to all the agents alive.

For a retiree participating the monthly plan, the consumption policy solves

$$V_t^m = \max \int_t^T e^{-\int_t^s \rho + m(x)dx} [u(c_s) + m(s)b(a_s)] ds \quad (11)$$

subject to

$$\dot{a}_t = r_t a_t + \text{tr}_t + p - c_t \quad (12)$$

$$a_t \geq 0. \quad (13)$$

Workers. After aging T_w , the individuals can start to work. Given interest rate r_t and wage w_t , the households consume optimally to maximize

$$V_t^w = \mathbb{E}_t \left[\int_t^T e^{-\int_t^s \rho + m(x) dx} [u(c_s) - \psi(s) + m(s)b(a_s)] ds \right] \quad (14)$$

subject to

$$\dot{a}_t = r_t a_t + (y_t - \tau(y_t)) + \text{tr}_t - c_t \quad (15)$$

$$a_t \geq 0 \quad (16)$$

$$y_t = w_t \exp(\bar{z}(t) + z_t) \quad (17)$$

where $\bar{z}(t)$ is the earning profile and $z_t \in \mathcal{Z} = \{z_0, \dots, z_k\}$ follows a continuous time Markov chain with intensity matrix Λ . We take $z_0 = -\infty$ for the unemployment state. The initial z at T_w is drawn randomly from the invariant distribution associated with Λ . The ψ is the disutility to work. The pension tax τ is of the form

$$\tau(y) = \tau_0 \min \{y, \bar{y}\} \quad (18)$$

which is truncated from above.

We allow the workers to retire at any age. Yet, the pension is only available for those retiring within the retirement window $[T_r - 5, T_r + 5]$, where T_r is the standard retirement age. The benefits ℓ and p are determined by the retirement age

$$p = (1 + 0.04 \times (\text{retire age} - T_r))\bar{p}, \quad \ell = (1 + 0.04 \times (\text{retire age} - T_r))\bar{\ell}. \quad (19)$$

$\bar{p}$ and $\bar{\ell}$ are the benefits of retiring at T_r . The retirees cannot return to work.

Youngs. The youngs are born with zero asset and cannot work before T_w . They are assumed to be hand to mouth so $c_t = \text{tr}_t$.

3.3. Firms

The production function is given by

$$Y_t = AK_t^\alpha L_t^{1-\alpha}. \quad (20)$$

Given r_t, w_t , the firms solve

$$\max_{K_t, L_t} AK_t^\alpha L_t^{1-\alpha} - r_t K_t - w_t L_t \quad (21)$$

where L_t is the aggregate effective labor. Taking the FOC and detrending by N_t gives

$$A\left(\frac{\hat{K}_t}{\hat{L}_t}\right)^\alpha = A\left(\frac{K_t}{L_t}\right)^\alpha = \frac{w_t}{1-\alpha}, \quad \text{and} \quad A\left(\frac{\hat{K}_t}{\hat{L}_t}\right)^{\alpha-1} = A\left(\frac{K_t}{L_t}\right)^{\alpha-1} = \frac{r_t}{\alpha}. \quad (22)$$

3.4. Pension System

The government collect pension taxes from the labor income and finance the pension payments ℓ and p . Let pension fund balance be B_t . The budget constraint is given by

$$dB_t = \left[r_t B_t + N_t \int \tau_w(w, z) w z d\mathfrak{M}_t - N_t \int \ell(s) \mathfrak{M}_t(\text{retiring with lumpsum}) - N_t \int p \mathfrak{M}_t(\text{retired with monthly}) \right] dt. \quad (23)$$

Detrending gives

$$d\hat{B}_t = \left[(r_t - \eta_t) \hat{B}_t + \int \tau_w(w, z) w z d\mathfrak{M}_t - \ell \mathfrak{M}(\text{retiring with lumpsum}) - p \mathfrak{M}(\text{retired with monthly}) \right] dt. \quad (24)$$

The reform occurs when the fund balance $\hat{B}_t$ becomes zero. There are two possible reforms: one is to let the pension system collapse and the other to raise the pension tax rate such that the pension system is just solvent in the long run. Agents have beliefs on the two possible reforms. Among all the agents, π proportion believe that the pension system would simply collapse in the future and $1 - \pi$ believe that the pension tax rate would be raised to prevent the system from insolvency.

3.5. Equilibrium

Given the initial distribution $\mathfrak{M}_0(da, ds, dz, dp, d\pi)$, pension fund balance $\hat{B}_0$, the recursive equilibrium consists of optimal consumption policy path c_t , the optimal retirement policy, the price paths $\{r_t, w_t\}$, the fund balance path $\hat{B}_t$ and the distribution $\mathfrak{M}_t$ such that

- (a) Given the prices r_t, w_t and $\hat{B}_t$, the optimal consumption policies and the retirement policies solves the individuals optimization problem.
- (b) Given the prices r_t and w_t , the firms optimize.
- (c) The prices are such that the asset market and the labor market clear:

$$\hat{K}_t = \int a d\mathfrak{M}_t + \hat{B}_t \quad (25)$$

$$\hat{L}_t = \int \exp(\bar{z}(s) + z) \mathbf{1}\{\text{working}\} d\mathfrak{M}_t. \quad (26)$$

- (d) The individuals optimal retirement decision R_t is consistent with the fund balance path $\hat{B}_t$.
- (e) The distribution solves the corresponding Kolmogorov forward equation.

4. Results

5. Conclusion

Appendix

A. Numerical Detail

References

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